Entropy-Based Analysis of Encryption

Essential Logical Model + Simple Worked Example

Goal: Take the same plaintext, encrypt it, calculate Shannon entropy before and after encryption, and compare the results.

1. Logical Flow

```
PLAINTEXT
   ↓
Character Frequency
   ↓
Probability of each character
   ↓
Plaintext Entropy H(X)
   ↓
ENCRYPTION
   ↓
CIPHERTEXT
   ↓
Character Frequency
   ↓
Ciphertext Entropy H(Y)
   ↓
Compare H(X) and H(Y)
   ↓
Display Results / Graph
```

2. Mathematical Model

Probability: p(x) = frequency of x / total characters
Shannon Entropy: H(X) = –Σ p(x) log■ p(x)
Entropy Change: ΔH = H(Y) – H(X)

3. Simple Worked Example

Plaintext: HELLO
Total characters = 5
Frequencies: H = 1, E = 1, L = 2, O = 1

Character	Frequency	Probability
H	1	1/5 = 0.20
E	1	1/5 = 0.20
L	2	2/5 = 0.40
O	1	1/5 = 0.20

Therefore:
H(HELLO) = –[3(0.20 log■0.20) + 0.40 log■0.40] ≈ 1.922 bits/character.

4. Encrypt the Same Text

For a simple demonstration, use a Caesar cipher with shift = 3:

```
HELLO
  ↓ Caesar shift +3
KHOOR
```

Ciphertext: KHOOR
Its character frequencies are K = 1, H = 1, O = 2, R = 1. The same entropy calculation is then applied to KHOOR.

For this particular Caesar-cipher example, H(KHOOR) ≈ 1.922 bits/character.

This happens because a Caesar cipher changes symbols but preserves their frequency pattern. Therefore, this example is useful for showing the model, but it also demonstrates why entropy alone cannot prove encryption strength.

5. Minimum Python Structure

```
def calculate_entropy(text):
    # count characters
    # calculate probabilities
    # calculate -p * log2(p)
    # return entropy
```

```
def encrypt(text, method):
    # encrypt plaintext
    # return ciphertext

plaintext = input("Enter plaintext: ")

H_plain = calculate_entropy(plaintext)
ciphertext = encrypt(plaintext, "selected_method")
H_cipher = calculate_entropy(ciphertext)

print("Ciphertext:", ciphertext)
print("Plaintext Entropy:", H_plain)
print("Ciphertext Entropy:", H_cipher)
print("Entropy Change:", H_cipher - H_plain)
```

6. Final Project Output

The final program should show the plaintext, encryption method, ciphertext, plaintext entropy, ciphertext entropy, and entropy change. For multiple encryption methods, the same plaintext should be used for every method and the entropy values can be displayed in a comparison table or bar graph.

Important: Entropy measures character-level unpredictability. It should not be treated by itself as proof that an encryption method is cryptographically secure.